

Dark-Activated Light Sensor Circuit Using LDR

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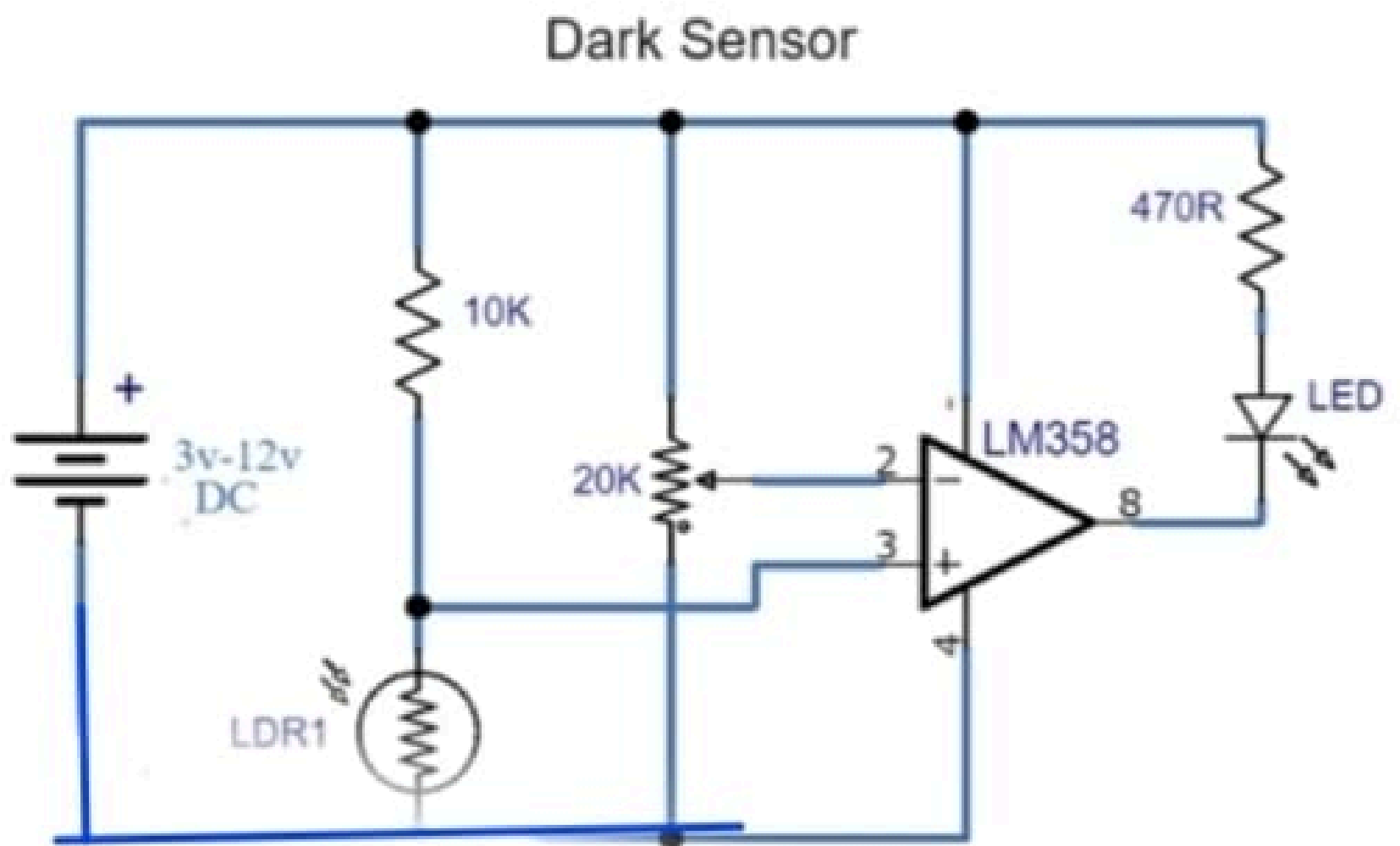
Project Description

This project demonstrates a simple dark-activated light sensor using a Light Dependent Resistor (LDR), a potentiometer, and the LM358 operational amplifier in comparator mode. The purpose of the circuit is to detect darkness and automatically switch ON an LED, making it suitable for night lamps, garden lights, or ambient light trigger. The LDR forms a voltage divider with a fixed resistor. As light decreases, the resistance of the LDR increases, which raises the input voltage to the comparator. A potentiometer sets the reference voltage to define the darkness threshold. When the LDR voltage exceeds the reference, the op-amp output goes high, turning on the Led.

Key features includes:

1. Adjustable Sensitivity – Using a potentiometer to set the light threshold.
2. Optional Dimming or Buzzer Alerts – Additional circuitry can drive brightness changes or alarms.
3. Debounce / Flicker Protection – A capacitor can suppress false triggering.
4. Relay Control – Can be extended to drive larger loads using a relay circuit.

Circuit Diagram



Component List

S.No	Component	Specification	Quantity	Purpose
1	LDR	5mm or 10mm	1	Senses ambient light
2	LM358 Op-Amp IC	Dual op-amp	1	Compares LDR voltage and reference
3	Potentiometer	10k Ω	1	Adjusts reference voltage
4	Fixed Resistor	10k Ω	1	Forms voltage divider with LDR
5	LED	Any colour (5mm)	1	Turns ON in darkness
6	Resistor for LED	330 Ω to 1k Ω	1	Limits LED current
7	Capacitor	0.1 μ F to 10 μ F	1	Reduces flicker/debounce
8	Relay Module	5V Single Channel Relay	1	For controlling AC loads (optional)
9	Transistor	BC547 or 2N2222	1	If driving relay from op-amp output
10	Diode	1N4007	1	Flyback protection for relay
11	Breadboard / Dot Board	–	1	For circuit assembly
12	Power Supply	5V or 9V Battery / Adapter	1	Provides circuit power
13	Connecting Wires	Male-to-Male Jumpers	As needed	For connections on breadboard

Conclusion

This project demonstrates how light-sensitive behavior can be implemented with simple components. Adjustable threshold and brightness make it practical for real-world applications like automatic lighting. Its low cost and scalability make it ideal for learning and prototyping.